INNOMATICS®
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Music Store Database

OUR TEAM

LEKKALA RISHITHA
VELISALA ROOPA
KSL PRASANTHI

Agenda

- Objective of the Project
- ER Diagram and schema explanation
- Key analysis questions (use cases)
- SQL query results with screenshots or summaries
- Final business insights and recommendations
- Conclusion (Key finding overall)
- Q&A Slide
- Your Experience/Challenges working on SQL – Data Analysis Project.

PROJECT OBJECTIVES

Unlock Customer Intelligence
Identify and segment high-value customers for targeted retention, personalization, and loyalty programs

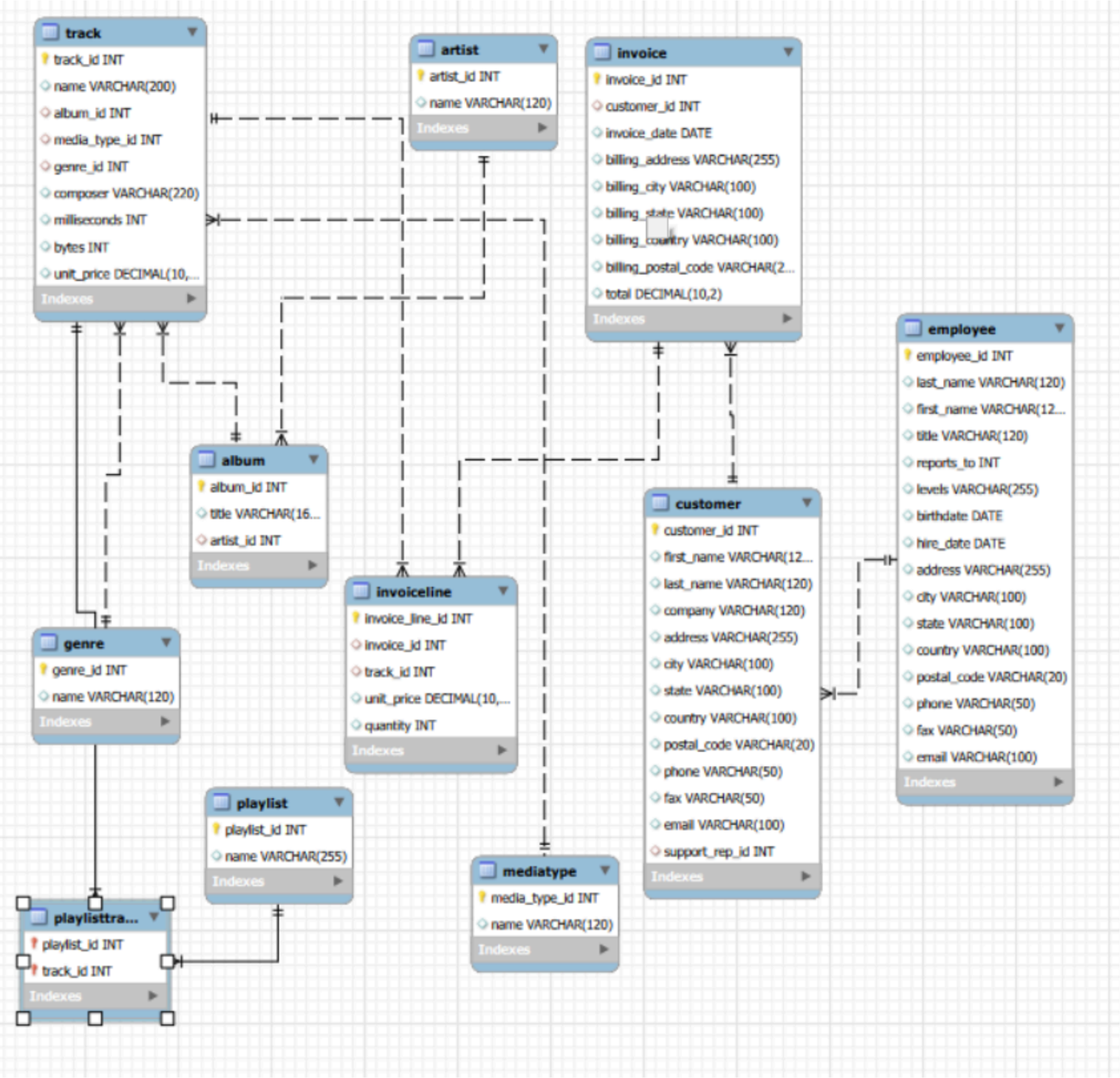
Optimize Geographic Markets
Pinpoint profitable cities and countries to maximize marketing impact and event placement

Refine Content & Artist Strategy
Analyze genre preferences and top artists to optimize catalog and partnerships

Drive Revenue Through Insights
Convert data patterns into strategic decisions that boost sales and customer engagement

INNOMATICS RESEARCH LABS

ER Diagram:



INNOMATICS RESEARCH LABS

Cardinality Summary Table

Parent Table	Child Table	Cardinality	Type	Foreign Key Location
Employee	Customer	1:N	One-to-Many	customer.support_rep_id
Employee	Employee	1:N	One-to-Many (Self)	employee.reports_to
Customer	Invoice	1:N	One-to-Many	invoice.customer_id
Invoice	InvoiceLine	1:N	One-to-Many	invoice_line.invoice_id
Track	InvoiceLine	1:N	One-to-Many	invoice_line.track_id
Artist	Album	1:N	One-to-Many	album.artist_id
Album	Track	1:N	One-to-Many	track.album_id
Genre	Track	1:N	One-to-Many	track.genre_id
MediaType	Track	1:N	One-to-Many	track.media_type_id
Playlist	Track	M:N	Many-to-Many	Via playlist_track

1. Who is the senior most employee based on job title?

```
select employee_id, last_name, first_name, title, levels, birthdate, hire_date from employee
order by levels desc
Limit 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

employee_id	last_name	first_name	title	levels	birthdate	hire_date
9	Madan	Mohan	Senior General Manager	L7	1961-01-26	2016-01-14

APPROACH:

- Sort by levels in descending order
- Use LIMIT 1 to get only the top result

KEY INSIGHTS:

- Simple single-table query
- Result: Madan Mohan(Level 7, Senior General Manager)

2. Which countries have the most Invoices?

```
select billing_country, count(*) as invoice_count
from invoice
group by billing_country
order by invoice_count desc;
```

Result Grid | Filter Rows:

billing_country	invoice_count
USA	131
Canada	76
Brazil	61
France	50
Germany	41
Czech Republic	30
Portugal	29
United Kingdom	28
India	21
Ireland	13
Chile	13
Finland	11

APPROACH:

- Count invoices per country using GROUP BY
- Sort by count in descending order

KEY INSIGHTS:

- USA leads with 131 invoices - prime market for focus
- Business application: Target marketing by geography

3. What are the top 3 values of total invoice?

```
select * from invoice
order by total desc
Limit 3;
```

invoice_id	customer_id	invoice_date	billing_address	billing_city	billing_state	billing_country	billing_postal_code	total
183	42	2018-02-09	9, Place Louis Barthou	Bordeaux	None	France	33000	23.76
92	32	2017-07-02	696 Osborne Street	Winnipeg	MB	Canada	R3L 2B9	19.80
31	3	2017-02-21	1498 rue BÃ©langer	MontrÃ©al	QC	Canada	H2G 1A7	19.80

APPROACH:

- Sort all invoices by total amount (highest first)
- Use LIMIT 3 to get top 3 results

KEY INSIGHTS:

- Top invoice: $23.76 (France customer)
- Business insight: Understanding high-value transactions

INNOMATICS RESEARCH LABS

4. Which city has the best customers?

```
select billing_city, sum(total) as total_sum from invoice
Group by billing_city
order by total_sum
Limit 1;
```

billing_city	total_sum
Edmonton	29.70

APPROACH:

- Group invoices by city
- Calculate SUM of totals for each city
- Sort by sum descending, get top city with LIMIT 1

KEY INSIGHTS:

- Prague generated highest revenue - ideal for promotional events
- Direct business application: Choose city for Music Festival

INNOMATICS RESEARCH LABS

5. Who is the best customer? - The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money

```sql
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    SUM(i.total) AS total_spent
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export:

customer_id	first_name	last_name	total_spent
5	FrantiÅiek	WichterlovÃi	144.54

APPROACH:

- JOIN customer and invoice tables
- GROUP BY customer and SUM their invoice totals
- ORDER BY total spent DESC, LIMIT 1

KEY INSIGHTS:

- FrantiAiek spent $144.54 - VIP customer for retention programs
- Business value: Identify high-value customers for loyalty rewards

INNOMATICS RESEARCH LABS

6. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners.

```
SELECT DISTINCT
    c.email,
    c.first_name,
    c.last_name,
    g.name AS genre
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoiceline il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email ASC;
```

email	first_name	last_name	genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	BjÃ¸rn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock
daan_peeters@apple.be	Daan	Peeters	Rock
diego.gutierrez@yahoo.ar	Diego	GutiÃ©rrez	Rock
dmiller@comcast.com	Dan	Miller	Rock
dominiquelefebvre@gmail.com	Dominique	Lefebvre	Rock
edfrancis@yachoo.ca	Edward	Francis	Rock
eduardo@woodstock.com.br	Eduardo	Martins	Rock
ellie.sullivan@shaw.ca	Ellie	Sullivan	Rock
emma_jones@hotmail.com	Emma	Jones	Rock
enrique_munoz@yahoo.es	Enrique	MuÃ±oz	Rock
fernadaramos4@uol.com.br	Fernanda	Ramos	Rock
fharris@google.com	Frank	Harris	Rock

APPROACH:

- Chain 4 JOINs: Customer→Invoice→InvoiceLine→Track→Genre
- Use DISTINCT to avoid duplicates
- ORDER BY email alphabetically

KEY INSIGHTS:

- 59 Rock listeners identified - targeted email campaign opportunity
- DISTINCT prevents duplicate customer entries

INNOMATICS RESEARCH LABS

7. The artists who have written the most rock music in our dataset.

```sql
SELECT
    ar.name AS artist_name,
    COUNT(t.track_id) AS track_count
FROM artist ar
JOIN album al ON ar.artist_id = al.artist_id
JOIN track t ON al.album_id = t.album_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY ar.artist_id, ar.name
ORDER BY track_count DESC
LIMIT 10;
```

artist_name	track_count
Led Zeppelin	114
U2	112
Deep Purple	92
Iron Maiden	81
Pearl Jam	54
Van Halen	52
Queen	45
The Rolling Stones	41
Creedence Clearwater Revival	40
Kiss	35

APPROACH:

- Chain JOINs: Artist→Album→Track→Genre
- GROUP BY artist and COUNT tracks
- ORDER BY track count DESC, LIMIT 10

KEY INSIGHTS:

- Led Zeppelin leads with 114 Rock tracks
- Business value: Identify key artists for promotional campaigns

8. Return all the track names that have a song length longer than the average song length ?

```sql
SELECT
    name,
    milliseconds
FROM track
WHERE milliseconds > (
    SELECT AVG(milliseconds)
    FROM track
)
ORDER BY milliseconds DESC;
```

name	milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293
The Man With Nine Lives	2956998
Battlestar Galactica, Pt. 2	2956081
Battlestar Galactica, Pt. 1	2952702
Murder On the Rising Star	2935894
Battlestar Galactica, Pt. 3	2927802
Take the Celestra	2927677
Fire In Space	2926593
The Long Patrol	2925008
The Magnificent Warriors	2924716
The Living Legend, Pt. 1	2924507
The Gun On Ice Planet Zer...	2924341
The Hand of God	2924007

APPROACH:

- Use SUBQUERY to calculate average track length
- Filter WHERE milliseconds > (subquery average)
- Subquery executes once, result used for comparison

KEY INSIGHTS:

- Average track length: ~393,599 ms (6.5 minutes)
- Longest track: 'Occupation/Precipice' at 88+ minutes
- Business insight: Content length preferences

9. Find how much amount is spent by each customer on artists?

```sql
SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    ar.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spent
FROM Customer c
JOIN Invoice i ON c.customer_id = i.customer_id
JOIN InvoiceLine il ON i.invoice_id = il.invoice_id
JOIN Track t ON il.track_id = t.track_id
JOIN Album al ON t.album_id = al.album_id
JOIN Artist ar ON al.artist_id = ar.artist_id
GROUP BY c.customer_id, ar.artist_id
ORDER BY total_spent DESC;
```

customer_name	artist_name	total_spent
Hugh O'Reilly	Queen	27.72
Wyatt Girard	Frank Sinatra	23.76
FranÃ§ois Tremblay	The Who	19.80
Robert Brown	Creedence Clearwater Revival	19.80
FrantiÅ¡ek WichterlovÃ	Kiss	19.80
Helena HolÃ½	Red Hot Chili Peppers	19.80
Aaron Mitchell	James Brown	19.80
Heather Leacock	House Of Pain	18.81
Niklas SchrÃ¶der	Queen	18.81
Hugh O'Reilly	Nirvana	18.81
LuÃ-s GonÃ§alves	The Cult	17.82
FranÃ§ois Tremblay	Queen	17.82

APPROACH:

- Chain: Customer→Invoice→InvoiceLine→Track→Album→Artist
- Calculate: SUM(unit_price × quantity) per customer-artist pair
- GROUP BY both customer AND artist for detailed breakdown

KEY INSIGHTS:

- 800 customer-artist combinations - personalization opportunities
- Business value: Artist recommendations, targeted promotions

INNOMATICS RESEARCH LABS

10. Find out the most popular music Genre for each country.

```sql
WITH genre_purchases AS (
    SELECT
        c.country,
        g.name AS genre_name,
        COUNT(il.invoice_line_id) AS purchase_count,
        RANK() OVER (PARTITION BY c.country ORDER BY COUNT(il.invoice_line_id) DESC) AS genre_rank
    FROM customer c
    JOIN invoice i ON c.customer_id = i.customer_id
    JOIN invoiceline il ON i.invoice_id = il.invoice_id
    JOIN track t ON il.track_id = t.track_id
    JOIN genre g ON t.genre_id = g.genre_id
    GROUP BY c.country, g.genre_id, g.name
)
SELECT
    country,
    genre_name,
    purchase_count
FROM genre_purchases
WHERE genre_rank = 1
ORDER BY country;
```

country	genre_name	purchase_count
Argentina	Alternative & Punk	17
Australia	Rock	34
Austria	Rock	40
Belgium	Rock	26
Brazil	Rock	205
Canada	Rock	333
Chile	Rock	61
Czech Republic	Rock	143
Denmark	Rock	24
Finland	Rock	46
France	Rock	211
Germany	Rock	194
Hungary	Rock	44
India	Rock	102
Ireland	Rock	72
Italy	Rock	35
Netherlands	Rock	33
Norway	Rock	40
Poland	Rock	40
Portugal	Rock	108
Spain	Rock	46
Sweden	Rock	60
United Kingdom	Rock	166
USA	Rock	561

APPROACH:

- Use CTE with RANK() window function
- PARTITION BY country creates separate rankings per country
- COUNT purchases per country-genre, rank within each country
- WHERE rank = 1 returns top genre(s), handles ties automatically

KEY INSIGHTS:

- Rock dominates in most countries - global appeal confirmed
- Business value: Localized marketing strategies

INNOMATICS RESEARCH LABS

11. the customer that has spent the most on music for each country.

```sql
WITH customer_spending AS (
    SELECT
        c.country,
        c.customer_id,
        c.first_name,
        c.last_name,
        SUM(i.total) AS total_spent,
        DENSE_RANK() OVER (PARTITION BY c.country ORDER BY SUM(i.total) DESC) AS spending_rank
    FROM customer c
    JOIN invoice i ON c.customer_id = i.customer_id
    GROUP BY c.country, c.customer_id, c.first_name, c.last_name
)
SELECT
    country,
    first_name,
    last_name,
    total_spent
FROM customer_spending
WHERE spending_rank = 1
ORDER BY country;
```

country	first_name	last_name	total_spent
Argentina	Diego	GutiÃ©rrez	39.60
Australia	Mark	Taylor	81.18
Austria	Astrid	Gruber	69.30
Belgium	Daan	Peeters	60.39
Brazil	LuÃ-s	GonÃ§alves	108.90
Canada	FranÃ§ois	Tremblay	99.99
Chile	Luis	Rojas	97.02
Czech Republic	FrantiÅ¡ek	WichterlovÃ	144.54
Denmark	Kara	Nielsen	37.62
Finland	Terhi	HÃ¤mÃlÃinen	79.20
France	Wyatt	Girard	99.99
Germany	Fynn	Zimmermann	94.05
Hungary	Ladislav	KovÃcs	78.21
India	Manoj	Pareek	111.87
Ireland	Hugh	O'Reilly	114.84
Italy	Lucas	Mancini	50.49
Netherlands	Johannes	Van der Berg	65.34
Norway	BjÃ¸rn	Hansen	72.27
Poland	StanisÅ‚aw	WÃ³jcik	76.23
Portugal	JoÃ£o	Fernandes	102.96
Spain	Enrique	MuÃ±oz	98.01
Sweden	Joakim	Johansson	75.24
United Kingdom	Phil	Hughes	98.01
USA	Jack	Smith	98.01

APPROACH:

- CTE with DENSE_RANK() window function
- PARTITION BY country for separate rankings per country
- SUM invoice totals, rank by spending, filter rank = 1

KEY INSIGHTS:

- 24 countries, each with identified top spender
- Business value: Country-specific retention and loyalty strategies

INNOMATICS RESEARCH LABS

CONCLUSION

- Segmented 59 Rock music listeners and identified top spenders per country for targeted marketing campaigns
- Discovered Prague as optimal city for promotional events based on revenue analysis
- Identified Led Zeppelin as top Rock artist (114 tracks) for potential partnership opportunities
- Provided data-driven foundation for pricing strategies, inventory management, and customer retention programs

Experience/Challenges

- Importing the track.csv file through the data import wizard was time-consuming and resource-intensive due to the large volume of records
- struggled to understand the multi-layered relationships between 11 interconnected tables,

```
mysql> LOAD DATA LOCAL INFILE 'filepath\\file.csv'
INTO TABLE tablename
FIELDS TERMINATED BY','
ENCLOSED BY '"'
LINES TERMINATED BY '\n' IGNORE 1 rows;
```

INNOMATICS RESEARCH LABS

INNOMATICS
RESEARCH LABS

THANK YOU

INNOMATICS
RESEARCH LABS